Lecture 8

Hopf algebras and quantum groups

Objectives. *Lecture 7 found braidings by solving equations. Where do they come from? For groups, the tensor product of representations comes from the diagonal $g \mapsto g \otimes g$, duals from $g \mapsto g^{-1}$, and the symmetry from the flip. This lecture asks which algebras H give a monoidal, rigid and braided* $\mathrm{Rep}(H)$*, and proves that bialgebras, Hopf algebras and quasitriangular Hopf algebras suffice. The quantum group $U_q(\mathfrak{sl}_2)$ then produces a braiding that is not a symmetry, and its matrix on two spins is, after a change of gauge, the braid limit of* Integrability*'s XXZ R-matrix. This is a side branch: Lecture 9 needs only that matrix.*

By the end of this lecture you will be able to:

- Define algebras, coalgebras, bialgebras and Hopf algebras in pictures, and prove that a bialgebra makes $\mathrm{Rep}(H)$ monoidal and an antipode makes it rigid.
- Prove that a universal R-matrix gives a braiding $c = P \circ R$, and tell R from c.
- State the reconstruction theorem, and show that Fibonacci and Ising are not representation categories of Hopf algebras.
- Compute with $U_q(\mathfrak{sl}_2)$ on $V_1 \otimes V_1$: the matrix of the braiding, the Hecke relation, and the dictionary to XXZ.

8.1 Bialgebras give monoidal representation categories

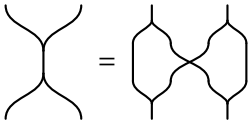
All modules here are finite-dimensional. The algebras acting on them may be infinite-dimensional, as for $U(\mathfrak{g})$ and U_q. The tensor product of linear maps is the ordinary one. An *algebra* is (H, μ, ι), with multiplication $\mu : H \otimes H \to H$ and unit $\iota : \mathbb{C} \to H$, associative and unital. A *coalgebra* is the same with arrows reversed: $(H, \Delta, \varepsilon_H)$, $\Delta : H \to H \otimes H$ coassociative and $\varepsilon_H : H \to \mathbb{C}$ a counit, $(\varepsilon_H \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon_H)\Delta$. In pictures, read upward, μ and Δ are a fusion and a splitting vertex:

$\mu =$  $\Delta =$ 

Kassel writes η for the unit and plain ε for the counit; here η and ε are taken.

Definition 8.1 (Bialgebra)**.** A *bialgebra* is an algebra and a coalgebra on the same H such that Δ and ε_H are algebra maps: $\Delta(xy) = \Delta(x)\Delta(y)$, $\Delta(1) = 1 \otimes 1$, $\varepsilon_H(xy) = \varepsilon_H(x)\varepsilon_H(y)$, $\varepsilon_H(1) = 1$.

The first condition is the *bialgebra law*; with P the flip of Vect (a plain crossing, Lecture 7) it reads

 (8.1)

that is, $\Delta \circ \mu = (\mu \otimes \mu) \circ (\mathrm{id} \otimes P \otimes \mathrm{id}) \circ (\Delta \otimes \Delta)$.

Theorem 8.2 (Bialgebras and monoidal categories)**.** *Let H be an algebra with algebra maps $\Delta\colon H \to H \otimes H$ and $\varepsilon_H\colon H \to \mathbb{C}$. If $(H, \Delta, \varepsilon_H)$ is a bialgebra, then* Rep(H) *is monoidal, with h acting on $V \otimes W$ by $\Delta(h)$ and on $\mathbb{C}$ by $\varepsilon_H(h)$, the associator and unitors those of* Vect*, and the forgetful functor to* Vect *strict monoidal. Conversely · sketched, for finite-dimensional H, if this recipe makes* Rep(H) *monoidal with these associators and unitors, then Δ is coassociative and ε_H a counit.*

Proof. Δ is an algebra map, so $\Delta(h)$ defines an action on $V \otimes W$; likewise ε_H on $\mathbb{C}$. If f and g intertwine, so does $f \otimes g$, since $(f \otimes g)\Delta(h) = \Delta(h)(f \otimes g)$ term by term. The associator of Vect, $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$, intertwines the actions $(\Delta \otimes \mathrm{id})\Delta(h)$ and $(\mathrm{id} \otimes \Delta)\Delta(h)$, which coassociativity makes equal; the unitors intertwine by the counit axiom. The pentagon and triangle hold because they hold in Vect. For the converse, apply the condition to $V = W = H$ itself, acting on itself, and evaluate at $1 \otimes 1 \otimes 1$ (Kassel Prop. XI.3.1). ∎

Example 8.3 (Three bialgebras)**.**

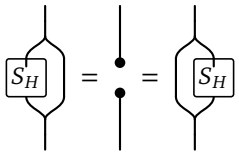
- The group algebra $\mathbb{C}[G]$ with $\Delta(g) = g \otimes g$, $\varepsilon_H(g) = 1$: g acts on $V \otimes W$ by $\rho_V(g) \otimes \rho_W(g)$, and Rep$(\mathbb{C}[G])$ is Rep(G) with its tensor product.
- The functions $\mathbb{C}^G$, with basis δ_g, pointwise product and $\Delta(\delta_g) = \sum_{hk=g} \delta_h \otimes \delta_k$: a module is a G-graded vector space, and Rep$(\mathbb{C}^G)$ is Vec_G; here $\varepsilon_H(\delta_g) = \delta_{g,1}$.
- The enveloping algebra $U(\mathfrak{g})$ with $\Delta(x) = x \otimes 1 + 1 \otimes x$ and $\varepsilon_H(x) = 0$ for $x \in \mathfrak{g}$: this is *QM-I*'s total angular momentum, $J_a = J_a^{(1)} \otimes 1 + 1 \otimes J_a^{(2)}$ (eq. (13.1)), adding angular momenta. (This J is angular momentum, not the structure map of a monoidal functor in Lecture 3.)

The coproduct is the rule for combining two systems. The diagonal $g \otimes g$ says a symmetry acts on both; the primitive $x \otimes 1 + 1 \otimes x$ says a generator, an infinitesimal symmetry, adds.

8.2 Antipodes give duals

Definition 8.4 (Hopf algebra)**.** A *Hopf algebra* is a bialgebra with an invertible linear map $S_H\colon H \to H$, the *antipode*, such that $\mu(S_H \otimes \mathrm{id})\Delta = \iota\,\varepsilon_H = \mu(\mathrm{id} \otimes S_H)\Delta$.

In pictures, with the counit a strand that ends and the unit one that begins:



(8.2)

Proposition 8.5 (Antipodes give duals · *proved given Kassel Thm. III.3.4*)**.** *If H is a Hopf algebra and V an H-module, then V^* with $(h \cdot f)(v) = f(S_H(h)v)$ is an H-module, and the evaluation and coevaluation of* Vect *are H-linear: a left dual. With S_H^{-1} in place of S_H, the same space is a right dual ${}^*V$. So* Rep(H) *is rigid.*

Proof. S_H reverses products, $S_H(xy) = S_H(y)S_H(x)$ (Kassel Thm. III.3.4), so the formula is an action. For $\mathrm{ev}(f \otimes v) = f(v)$, write $\Delta(h) = \sum h_{(1)} \otimes h_{(2)}$; then $\mathrm{ev}(\Delta(h)(f \otimes v)) = \sum f(S_H(h_{(1)})h_{(2)}v) = \varepsilon_H(h)f(v)$, by the antipode axiom: h acts on $\mathbb{C}$ by $\varepsilon_H(h)$, as it should. The coevaluation is checked in the same way with the other half of the axiom (Kassel

Props. III.5.2–5.3). The right dual uses S_H^{-1}, which satisfies the antipode identities for the opposite products, and is checked the same way with the evaluation $V \otimes {}^*V \to \mathbb{C}$ (EGNO Cor. 5.3.7; this is where invertibility is used). ∎

For $\mathbb{C}[G]$, $S_H(g) = g^{-1}$, and V^* is the contragredient representation of Lecture 4. For $U(\mathfrak{g})$, $S_H(x) = -x$: x acts on V^* by $-x^T$, which is why the dual of a spin has the opposite weights. For $\mathbb{C}^G$, $S_H(\delta_g) = \delta_{g^{-1}}$, and the dual of a vector of degree g has degree g^{-1}, as in Vec_G.

8.3 Quasitriangularity gives a braiding

The flip $P\colon v \otimes w \mapsto w \otimes v$ is H-linear for $\mathbb{C}[G]$ and $U(\mathfrak{g})$, because their coproducts are symmetric. In general P intertwines Δ with the opposite coproduct $\Delta^{\mathrm{op}} = P\Delta$, and a braiding needs a correction.

Definition 8.6 (Quasitriangular Hopf algebra)**.** A *universal R-matrix* is an invertible $R \in H \otimes H$ with

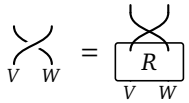
$$R\,\Delta(x)\,R^{-1} = \Delta^{\mathrm{op}}(x) \quad (x \in H),$$
$$(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23},$$
$$(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12},$$

where $R_{12} = R \otimes 1$, and so on. Kassel calls (H, R) *braided* (Def. VIII.2.2).

Theorem 8.7 (R gives a braiding)**.** *For a universal R-matrix, $c_{V,W} = P \circ R_{V,W}\colon V \otimes W \to W \otimes V$ is a braiding on* $\mathrm{Rep}(H)$*, and R satisfies*

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

In pictures, the crossing is a box followed by the plain flip:



(8.3)

Proof. c is H-linear: $c\,\Delta(x) = P\,R\,\Delta(x) = P\,\Delta^{\mathrm{op}}(x)\,R = \Delta(x)\,P\,R$, since $P\Delta^{\mathrm{op}}(x) = \Delta(x)P$. It is natural because R acts through the actions, which intertwiners commute with. For the second hexagon of Definition 7.2, R acts on $(U \otimes V) \otimes W$ by $(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23}$, and $P_{U\otimes V,W} = (P_{U,W} \otimes \mathrm{id})(\mathrm{id} \otimes P_{V,W})$; moving $\mathrm{id} \otimes P_{V,W}$ past R_{13} turns it into R_{12}, so $c_{U\otimes V,W} = (c_{U,W} \otimes \mathrm{id})(\mathrm{id} \otimes c_{V,W})$. The first follows in the same way from $(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12}$. Finally the braid relation of Theorem 7.4 for $c = P \circ R$ is, after moving the flips to one side, the displayed equation (Kassel Prop. VIII.3.1, Thm. VIII.2.4). ∎

So R is not the braiding: $c = P \circ R$, Kassel's $\tau \circ R$. This is the *Integrability* equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$, now with no spectral parameter, and c satisfies the braid form of Lecture 7. For $\mathbb{C}[G]$ and $U(\mathfrak{g})$, $R = 1 \otimes 1$ and $c = P$: symmetric. But the same Hopf algebra can carry another R. On $\mathbb{C}[\mathbb{Z}_2] = \mathrm{span}(1, g)$,

$$R = \tfrac{1}{2}\,(1 \otimes 1 + 1 \otimes g + g \otimes 1 - g \otimes g)$$

satisfies the axioms and acts on $V_a \otimes V_b$, where g acts by $(-1)^a$, by $(-1)^{ab}$: the braiding of sVec (Lecture 7). A braiding is structure, and here it is the choice of R.

8.4 Reconstruction, and what it forbids

A *fibre functor* is a faithful $\mathbb{C}$-linear monoidal functor to Vect, like the forgetful functor of $\mathrm{Rep}(H)$ (EGNO Def. 5.1.1; on a semisimple category the exactness EGNO asks for is automatic).

> **Theorem 8.8** (Reconstruction · *cited*). *A fusion category with a fibre functor is equivalent to* $\mathrm{Rep}(H)$ *for a finite-dimensional semisimple Hopf algebra* H*, built from the endomorphisms of the functor (EGNO Thm. 5.3.12, applied to fusion categories).*

A fibre functor preserves Frobenius–Perron dimensions (EGNO Prop. 4.5.7), and in Vect they are dimensions, integers. Fibonacci has $\mathrm{FPdim}(\tau) = \varphi$ and Ising $\mathrm{FPdim}(\sigma) = \sqrt{2}$, so neither has a fibre functor: no Hopf algebra has Fibonacci or Ising as its representations. Integer FPdims are necessary, not sufficient: Vec_G^{ω} with ω not a coboundary has every FPdim 1 and no fibre functor (EGNO Ex. 5.1.3), since a fibre functor would trivialise ω. The obstruction is FPdim, not the pivotal d: Vec_G for $G = \mathbb{Z}_3$ has the fibre functor that forgets the grading, yet a pivotal structure with $d = e^{2\pi i/3}$ (Lecture 4). The Fibonacci category does come from a quantum group, but as a subcategory of a quotient of its representations at a root of unity, the subject of Lecture 9.

8.5 $U_q(\mathfrak{sl}_2)$ and the XXZ dictionary

Fix $q \in \mathbb{C}^\times$ that is not a root of unity. The algebra $U_q(\mathfrak{sl}_2)$ is generated by E, F, $K^{\pm 1}$ with

$$KEK^{-1} = q^2 E, \qquad KFK^{-1} = q^{-2} F,$$
$$EF - FE = \frac{K - K^{-1}}{q - q^{-1}},$$

and it is a Hopf algebra with

$$\Delta(E) = K \otimes E + E \otimes 1,$$
$$\Delta(F) = 1 \otimes F + F \otimes K^{-1},$$
$$\Delta(K) = K \otimes K,$$
$$S_H(E) = -K^{-1}E, \quad S_H(F) = -FK,$$
$$S_H(K) = K^{-1},$$

and $\varepsilon_H(E) = \varepsilon_H(F) = 0$, $\varepsilon_H(K) = 1$ · *cited* (Kassel Prop. VII.1.1). This Δ is the opposite of Kassel's and EGNO's (Thm. 5.6.2), and S_H is the inverse of Kassel's antipode, as it must be for the opposite coproduct; the course uses it throughout. We use only the modules $V_1^{\otimes n}$ and their direct summands: on them K acts by integer powers of q, and the braiding below exists. (On the one-dimensional module with $K = -1$ and $E = F = 0$, which also satisfies the relations, no braiding is compatible with V_1.) On $V_1 = \mathbb{C}^2$, with basis $e_1 = \uparrow$, $e_2 = \downarrow$,

$$K = \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix},$$
$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

As $q \to 1$, E, F and $(K - K^{-1})/(q - q^{-1})$ become the e, f, h of $\mathfrak{sl}_2$ (the E, F, H of *Symmetries*, L12), and Δ becomes the primitive coproduct. This is a limit, not substitution at $q = 1$: the presentation above has denominator $q - q^{-1}$.

Theorem 8.9 (The braiding on two spins · *cited; proved on* $V_1 \otimes V_1$)**.** *The formal quantum group* $U_\hbar(\mathfrak{sl}_2)$*, over* $\mathbb{C}[[\hbar]]$ *with* $q = e^{\hbar/2}$*, has a universal R-matrix (Kassel Thm. XVII.4.2, his h), and with its legs exchanged it serves our* Δ *(his Lemma VIII.2.3); it acts on the modules* $V_1^{\otimes n}$ *and their summands, giving a braiding there (EGNO Ex. 8.3.13). On* $V_1 \otimes V_1$*, with* $q = A^2$*, it is* $R = \operatorname{diag}(A, A^{-1}, A^{-1}, A)\,(1 + (q - q^{-1})F \otimes E)$*, and the braiding is* $c = P \circ R = A^{-1}B_q$*, with*

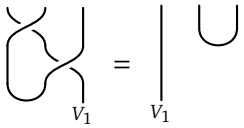
$$B_q = \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & q - q^{-1} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}$$

in the basis $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$.

Proof on $V_1 \otimes V_1$. Kassel's series, legs exchanged, is $q^{h\otimes h/2} \sum_n c_n F^n \otimes E^n$ with $c_0 = 1$, $c_1 = q - q^{-1}$; on V_1, $E^2 = F^2 = 0$, so only $n \le 1$ survives, and $q^{h\otimes h/2} = \operatorname{diag}(A, A^{-1}, A^{-1}, A)$ since $h = \operatorname{diag}(1, -1)$. Direct computation then gives $R\,\Delta(x)R^{-1} = \Delta^{\mathrm{op}}(x)$ for $x = E, F, K$, the two coproduct conditions on $V_1^{\otimes 3}$, and $PR = A^{-1}B_q$ (`scripts/check-l8-algebra.py`). ∎

B_q satisfies the braid relation and the *Hecke relation* $(B_q - q)(B_q + q^{-1}) = 0$: its eigenvalues are q on the triplet and $-q^{-1}$ on the *q-singlet* $v = e_1 \otimes e_2 - q\,e_2 \otimes e_1$, the cup of Lecture 9. So c has eigenvalues A and $-A^{-3}$, and $c^2 \neq 1$: a braiding that is not a symmetry. At $q = 1$, $B_q = P$. The q-singlet is invariant: $\Delta(E)v = \Delta(F)v = 0$ and $\Delta(K)v = v$, so $v\colon \mathbb{C} \to V_1 \otimes V_1$ is a module map, and V_1 is its own dual. The operator $U = B_q - q$, the projection onto the singlet up to a factor, satisfies $U^2 = -(q + q^{-1})U$ (Sheet 8); it is the generator of the Temperley–Lieb algebra of Lecture 9.

Naturality of the braiding with the cup $v\colon \mathbb{C} \to V_1 \otimes V_1$ demands that a strand pass the cup as if it were not there, $c_1 c_2 (v \otimes \mathrm{id}) = \mathrm{id} \otimes v$:



(8.4)

Write $e_{ijk} = e_i \otimes e_j \otimes e_k$. Applying the displayed matrix B_q on the adjacent factors gives

$$\begin{aligned} B_1B_2(v \otimes e_1) &= qe_{112} - q^2e_{121} \\ &= q(e_1 \otimes v), \\ B_1B_2(v \otimes e_2) &= qe_{212} - q^2e_{221} \\ &= q(e_2 \otimes v). \end{aligned}$$

Thus for a scalar $z \in \mathbb{C}^\times$ and a crossing $c = zB_q$, cup naturality requires $z^2 q = 1$. Choosing a root A with $A^2 = q$ gives $z = A^{-1}$; the other sign corresponds to the other root. The Hecke-normalised B_q solves the braid relation, but it fails this naturality condition.

Integrability's XXZ chain has the trigonometric R-matrix of its eq. (15.22), with spectral parameter λ and anisotropy $\Delta = \cos\eta$. It contains B_q as the end of its spectral line:

Integrability	this course
anisotropy $\Delta = \cos\eta$	$q = e^{i\eta} = A^2$, $\Delta = (q + q^{-1})/2$
$R(\lambda)$, after the gauge $\operatorname{diag}(e^{\lambda/2}, e^{-\lambda/2})$ on factor 1	the same, $\tilde{R}(\lambda)$
$\lim_{\lambda\to-\infty}(-2e^{\lambda})P\tilde{R}(\lambda)$	B_q, the over-crossing up to A
$\lim_{\lambda\to+\infty}(2e^{-\lambda})P\tilde{R}(\lambda)$	B_q^{-1}, the under-crossing up to A^{-1}
$\eta \to 0$	$q \to 1$, $B_q \to P$

The derivation is on Sheet 8, after the *Quantum Computing* survey. The two ends of the spectral line are the two crossings.

Summary.

- *Handed on:* a coproduct combines two systems; a bialgebra makes Rep(H) monoidal (Theorem 8.2), an antipode makes it rigid (Proposition 8.5), a universal R-matrix makes it braided, with $c = P \circ R$ (Theorem 8.7).
- *Reconstruction:* a fibre functor makes a fusion category Rep(H) (Theorem 8.8); integer FPdims are necessary, so Fibonacci and Ising are not Rep(H).
- *The quantum group:* on two spins, $c = A^{-1}B_q$ with $(B_q - q)(B_q + q^{-1}) = 0$ (Theorem 8.9); B_q is, after a change of gauge, the braid limit of the XXZ R-matrix.
- *Next:* Lecture 9 turns B_q into the Temperley–Lieb and Kauffman-bracket calculus, adds the twist, and builds $\mathrm{SU}(2)_k$.

References

Kassel, *Quantum Groups*: algebras, coalgebras, bialgebras and antipodes (Thm. III.3.4), §§III.1–III.3, pp. 39–52; modules and their duals, Props. III.5.1–III.5.3, pp. 58–60; $U(\mathfrak{g})$, Prop. V.2.4, p. 97; $U_q(\mathfrak{sl}_2)$, Def. VI.1.1, p. 122, and its Hopf structure, §VII.1, p. 140; braided bialgebras, Def. VIII.2.2, Lemma VIII.2.3 and Thm. VIII.2.4, pp. 173–175, and their braidings, Prop. VIII.3.1, p. 178; Prop. XI.3.1, p. 285; the universal R, Thm. XVII.4.2, p. 413. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: fibre functors, Def. 5.1.1, p. 91; reconstruction, Thm. 5.3.12, p. 96; duals, Cor. 5.3.7, p. 95; $U_q(\mathfrak{sl}_2)$, §5.6, p. 101; quasitriangular Hopf algebras, Def. 8.3.1, p. 198; the $U_q(\mathfrak{sl}_2)$ braiding, Ex. 8.3.13, p. 200; FPdim preserved, Prop. 4.5.7, p. 73. Heunen and Vicary, *Categories for Quantum Theory*: monoids and comonoids, Defs. 4.1, 4.3, pp. 128–129; Hopf algebras, Defs. 6.20, 6.31, pp. 205–210. Majid, *A Quantum Groups Primer*, Defs. 1.1–1.3 and 5.1, pp. 2–4, 29.